

Regularisation and early stopping

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Training and validation curves

Without opening the old lesson, explain one key idea from Training and validation curves. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Batching and epochs

Without opening the old lesson, explain one key idea from Batching and epochs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Gradients with a numerical check

Without opening the old lesson, explain one key idea from Gradients with a numerical check. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the next update for Regularisation and early stopping. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Regularisation and early stopping. Proposed explanation: “Zero prediction error means no regularised update”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Transfer both rules independently

For a new objective $J(a) = (a - 2)^2 + a^2$, gradient $4a - 4$, start $a = 0$ and take one step at rate 0.25. Give new a , prediction error term, penalty and total J . Separately apply patience 2 with strict improvement to validation values $[3, 2, 2, 2, 1]$ at epochs 0 to 4; report the stop and restored epochs.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Separate the objective terms

At $v = 2$ and $\lambda = 0.5$ in the lesson, compute prediction MSE, penalty, total objective and total gradient. Does zero training MSE make this gradient zero?

My reasoning and evidence

6. W2 Trace the next update

Start $v = 1.6$ with $\lambda = 0.5$ and rate 0.1. Calculate both gradient contributions, new v and new prediction MSE.

My reasoning and evidence

7. W3 Apply patience without peeking

Validation MSE at epochs 0 to 7 is [4, 2, 1.5, 1.6, 1.4, 1.45, 1.46, 1.3]. Apply the lesson policy with patience 2. Give the stop epoch, restored epoch and why the listed epoch 7 cannot change this decision.

My reasoning and evidence

8. W4 Transfer both rules independently

For a new objective $J(a) = (a - 2)^2 + a^2$, gradient $4a - 4$, start $a = 0$ and take one step at rate 0.25. Give new a , prediction error term, penalty and total J . Separately apply patience 2 with strict improvement to validation values [3, 2, 2, 2, 1] at epochs 0 to 4; report the stop and restored epochs.

My reasoning and evidence

9. W5 Compare penalties fairly

A team compares $\lambda = 0$ and $\lambda = 0.5$ by choosing whichever has the smaller penalised training objective. Explain why this does not establish the better prediction model and name a suitable development comparison.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled regularisation and early stopping evidence sheet to Image-pattern Network Investigation. Show how you can separate a weight penalty from prediction error. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Regularisation and early stopping

Separate a regularisation penalty from prediction loss and an early-stopping rule.

Toy objective $J(a) = (a-3)^2 + 2a^2$; start $a = 0$; gradient $6a-6$; rate $1/6$. Separate validation losses: [5, 3, 3, 4], patience 1, no ties as improvement.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Regularisation and early stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Regularised update

At 0 gradient -6; update gives $a = 1$.



2. Split objective

At 1 prediction term 4, penalty 2, total 6; at 0 total 9.



3. Early-stop separately

Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

CHANGE ONE THING

Change patience from 1 to 2 under the same tie rule.

Explain the changed result using the matching lesson.

Why should the penalty value not be reported as validation error? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate a regularisation penalty from prediction loss and an early-stopping rule.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3.

1. Split objective
2. Early-stop separately
3. Regularised update

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Early-stop separately Best validation checkpoint is 2 with loss 3; checkpoint 3 is no improvement, so stop at 3. Regularisation adds a parameter penalty to the training objective. Here it discourages large a , so the objective balances prediction error against that penalty. Early stopping is a different decision based on the supplied validation sequence. A tied validation loss is not improvement under this policy; stop when the patience allowance is used and restore the best checkpoint.

Practice 2: explain the deciding evidence

Changed-case practice: Change patience from 1 to 2 under the same tie rule. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. Stop at checkpoint 4; still restore checkpoint 2.

3. This one example guarantees correct results for every future input.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Stop at checkpoint 4; still restore checkpoint 2. Checkpoints 3 and 4 are two consecutive non-improvements; a later stop does not redefine the best checkpoint.

Repair a missing building block

Review Training and validation curves (ai-5/15). Goal: Compute comparable training and validation losses at saved checkpoints. Short practice: Can the original epoch 3 training loss 1 justify claiming best generalisation? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Batching and epochs (ai-5/13). Goal: Distinguish an example a batch an update and an epoch. Short practice: Why do the single-example orders finish at different weights? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Gradients with a numerical check (ai-5/10). Goal: Estimate one partial derivative while other parameters stay fixed. Short practice: Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
